

Assignment-1.R

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#####ASSIGNMENT-1#####

#Problem-1

```
library(alr4)
```

```
## Loading required package: car
```

```
## Loading required package: carData
```

```
## Loading required package: effects
```

```
## lattice theme set by effectsTheme()
```

```
## See ?effects theme for details.
```

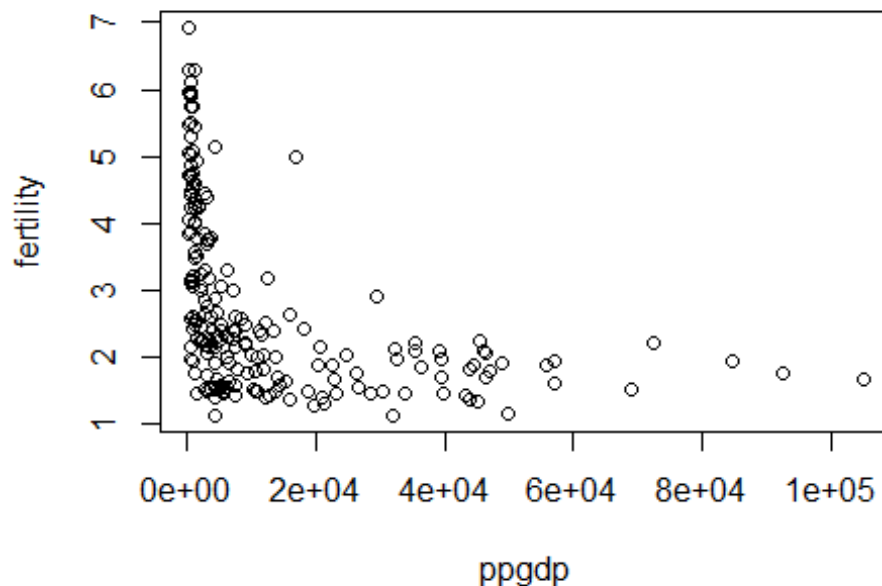
```
head(UN11)
```

##	region	group	fertility	ppgdp	lifeExpF	pctUrban
## Afghanistan	Asia	other	5.968	499.0	49.49	23
## Albania	Europe	other	1.525	3677.2	80.40	53
## Algeria	Africa	africa	2.142	4473.0	75.00	67
## Angola	Africa	africa	5.135	4321.9	53.17	59
## Anguilla	Caribbean	other	2.000	13750.1	81.10	100
## Argentina	Latin Amer	other	2.172	9162.1	79.89	93

A) Here, the predictor(X) is ppgdp(gross national product per person) and the response(Y) is fertility.

B) Plotting scatterplot for fertility and ppgdp

```
plot(fertility~ppgdp,data = UN11)
```



As you can see in the figure that the resulting scatterplot is not distributed properly, the mean is nonlinear and the variance is not constant also the value in the X is not the numerical form. so, we use the logarithm function for both X and Y.

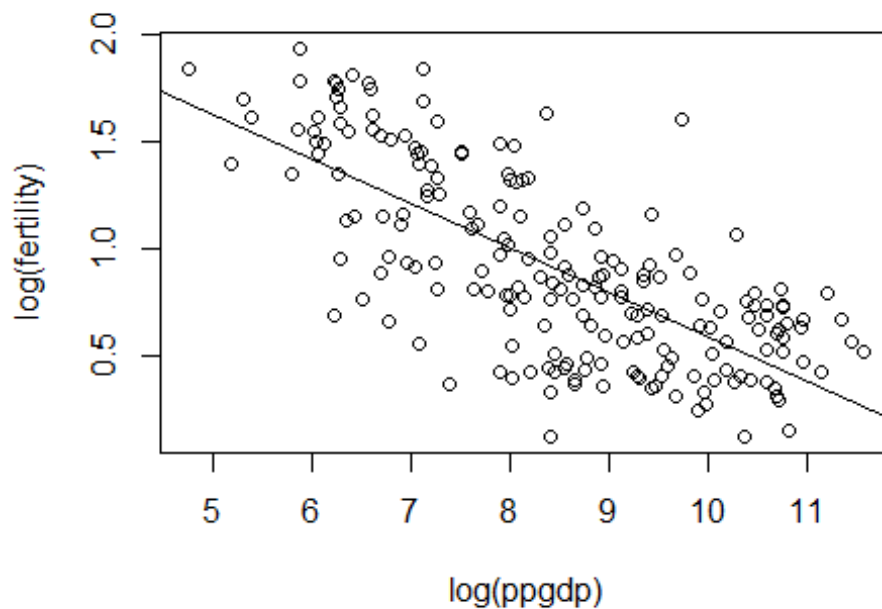
C) Using Log:

```
plot(log(fertility)~log(ppgdp),data = UN11)
lm(log(fertility)~log(ppgdp),data = UN11)

##
## Call:
## lm(formula = log(fertility) ~ log(ppgdp), data = UN11)
##
## Coefficients:
## (Intercept)    log(ppgdp)
##      2.6655      -0.2071
```

As we see, the slope is negative, so the mean function is downward sloping.

```
abline(lm(log(fertility)~log(ppgdp),data = UN11))
```



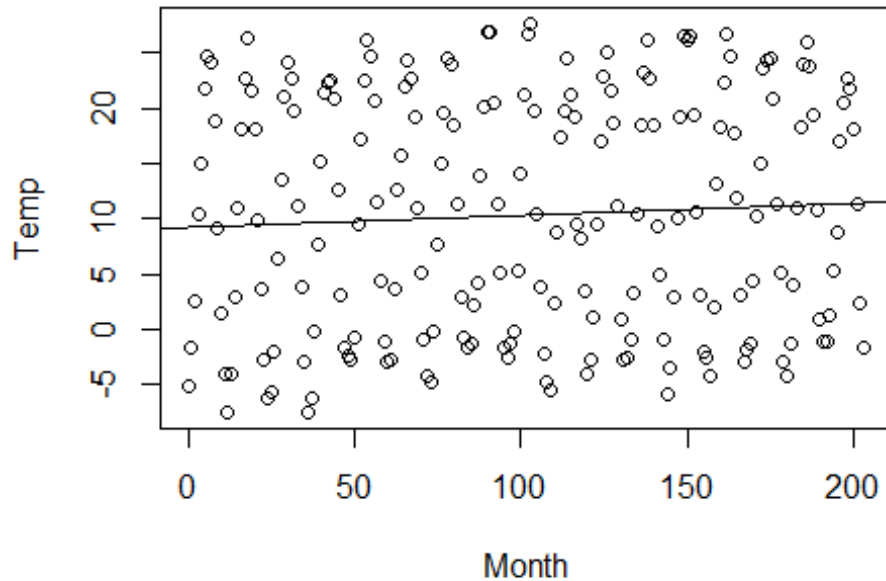
When compared to the previous graph this graph is good. The mean in this graph is linear and the variance is constant. As we have in the question that when we use the log function only the Values of the axis change, not the shape. So, the relation between ppgdp and fertility is linear and fit.

```
#problem-3
plot(Temp~Month,Mitchell)
lm(Temp~Month,Mitchell)

##
## Call:
## lm(formula = Temp ~ Month, data = Mitchell)
##
## Coefficients:
## (Intercept)      Month
##    9.27211    0.01025

abline(lm(Temp~Month,Mitchell))
```

A) we plot a scatterplot with the month as a predictor and temp as a response.



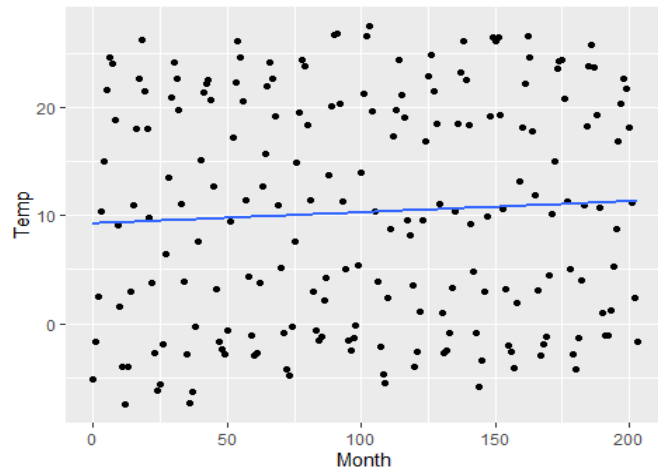
According to the text when the result line is the horizontal line and the slope is Zero then the graph is said to be a **null plot**.

We again plot this with ggplot:-

```
library(tidyverse)

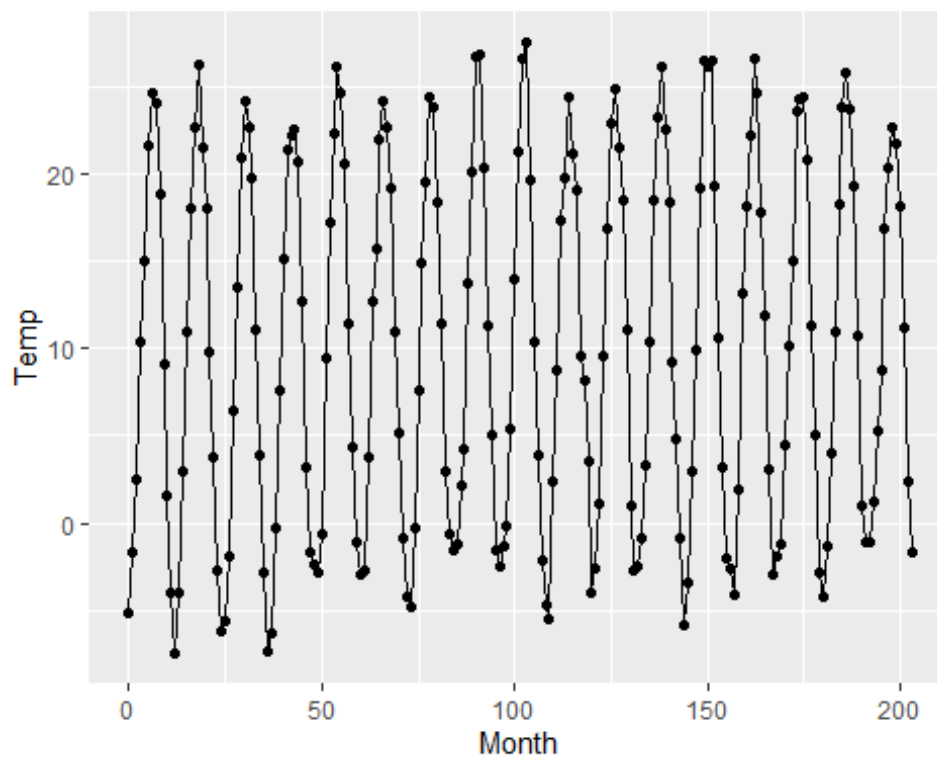
ggplot(aes(x = Month, y = Temp), data = Mitchell) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE)

## `geom_smooth()` using formula 'y ~ x'
```



B) I couldn't resize the graph but I used the ggplot line function to finish the analysis.

```
ggplot(aes(x = Month, y = Temp), data = Mitchell) +  
  geom_point() +  
  geom_line()
```



By reading the graph for every 6 months the temperature is high may be because of summer and the remaining 6 months of winter with less temperature.

#problem-4

```
head(oldfaith)
```

```
##   Duration Interval
## 1      216       79
## 2      108       54
## 3      200       74
## 4      137       62
## 5      272       85
## 6      173       55
```

```
plot(Interval~Duration,oldfaith)
```

```
lm(Interval~Duration,oldfaith)
```

```
##
```

```
## Call:
```

```
## lm(formula = Interval ~ Duration, data = oldfaith)
```

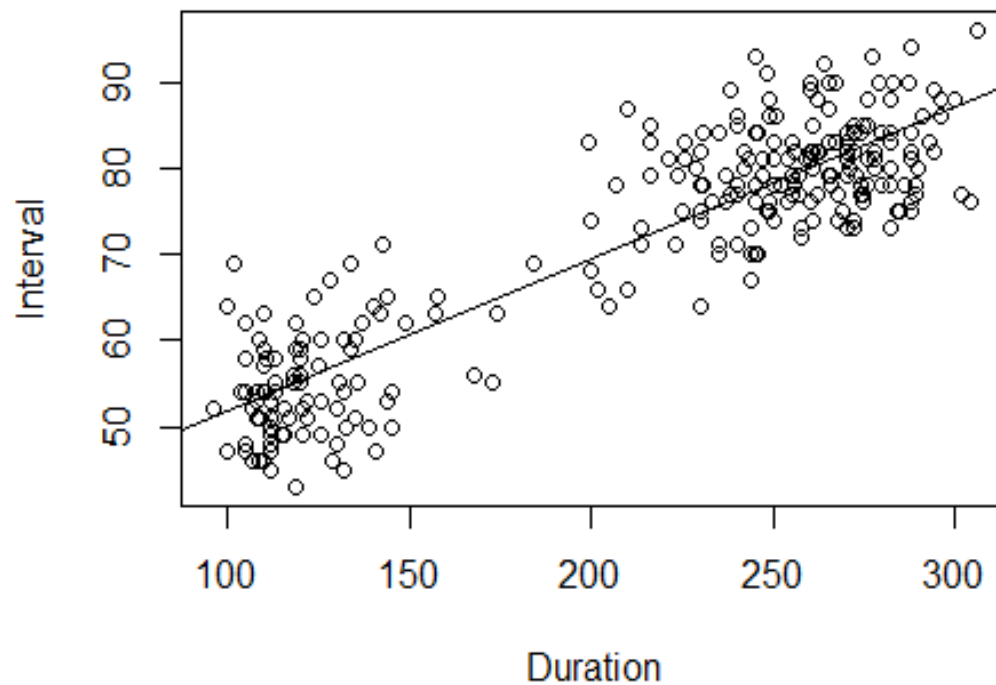
```
##
```

```
## Coefficients:
```

```
## (Intercept)      Duration
```

```
##      33.9878       0.1769
```

```
abline(lm(Interval~Duration,oldfaith))
```



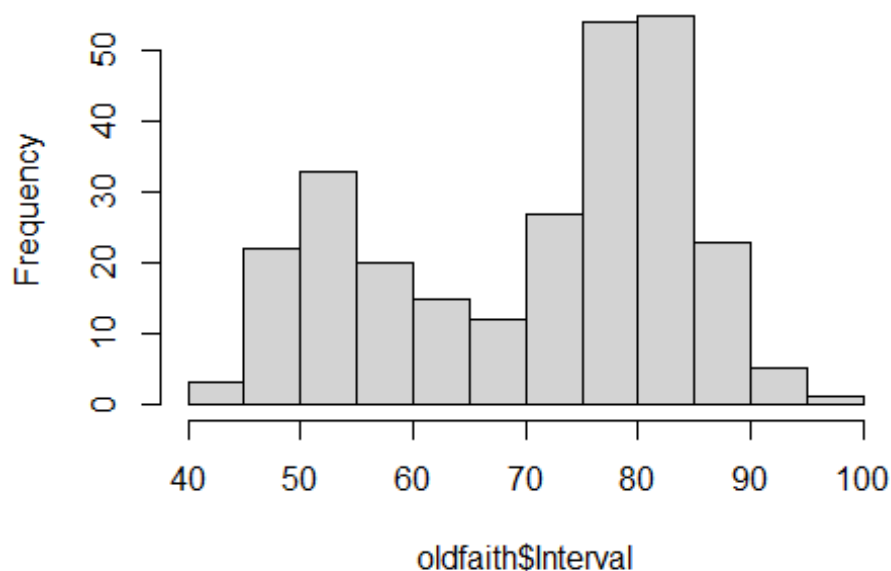
Conditional Distribution:-

oldfaith\$Interval

```
## [1] 79 54 74 62 85 55 88 85 51 85 54 84 78 47 83 52 62 84 52 79 51 47 78
69 74
## [26] 83 55 76 78 79 73 77 66 80 74 52 48 80 59 90 80 58 84 58 73 83 64 53
82 59
## [51] 75 90 54 80 54 83 71 64 77 81 59 84 48 82 60 92 78 78 65 73 82 56 79
71 62
## [76] 76 60 78 76 83 75 82 70 65 73 88 76 80 48 86 60 90 50 78 63 72 84 75
51 82
## [101] 62 88 49 83 81 47 84 52 86 81 75 89 79 69 81 50 85 51 87 63 67 77 56
88 81
## [126] 82 55 90 45 83 56 89 46 82 51 86 53 79 81 69 82 77 76 59 80 49 96 53
77 77
## [151] 65 81 71 70 81 93 53 89 45 86 58 78 66 76 63 88 52 93 49 57 77 68 81
81 73
## [176] 50 85 74 55 77 83 83 51 78 84 46 83 55 81 57 76 84 77 81 87 77 51 78
60 82
## [201] 91 53 78 46 77 84 49 83 71 80 49 75 64 76 53 94 55 76 50 82 54 75 78
79 78
## [226] 78 70 79 70 54 86 50 90 54 54 77 79 64 75 47 86 63 85 82 57 82 67 74
54 83
## [251] 73 73 88 80 71 83 56 79 78 84 58 83 43 60 75 81 46 90 46 74
```

hist(oldfaith\$Interval)

Histogram of oldfaith\$Interval



```
summary(oldfaith$Interval)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  43.00   58.00   76.00   71.11   82.00   96.00
```

Conclusion:-

According to the scatterplot and the summary, when the duration of the eruption is less the time interval of the next eruption is less. For example, in the figure, if the duration of the eruption is 100sec then the time taken for the next interval is 53 minutes. If the duration of the eruption is more the time interval of the next eruption is more.

#problem-2

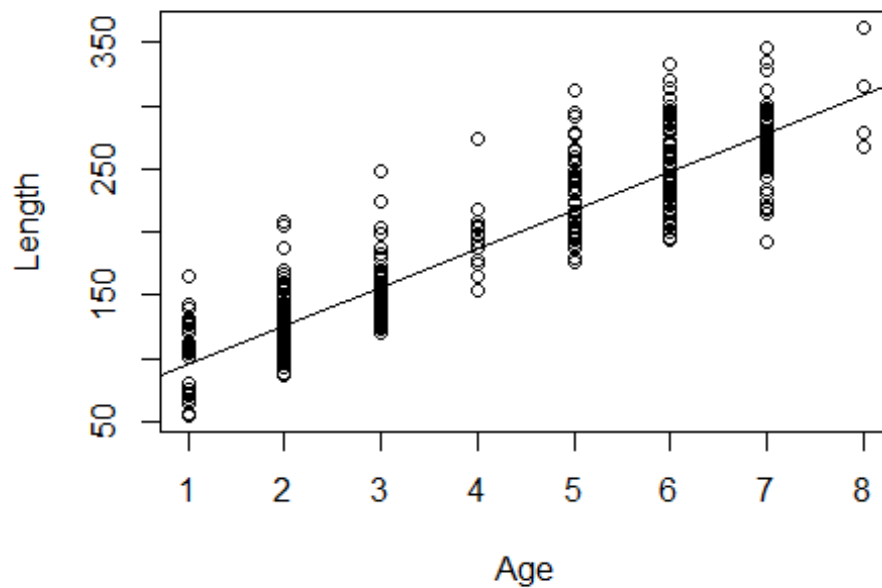
```
head(wblake)
```

```
##      Age Length    Scale
## 1     1      71 1.90606
## 2     1      64 1.87707
## 3     1      57 1.09736
## 4     1      68 1.33108
## 5     1      72 1.59283
## 6     1      80 1.91602
```

```
plot(Length~Age,wblake)
lm(Length~Age,wblake)
```

```
##
## Call:
## lm(formula = Length ~ Age, data = wblake)
##
## Coefficients:
## (Intercept)          Age
##      65.53         30.32
```

```
abline(lm(Length~Age,wblake))
```

The mean is linear and the variances are constant.

```
var(wblake$Length[wblake$Age==1])
```

```
## [1] 808.2312
```

```
var(wblake$Length[wblake$Age==2])
```

```
## [1] 697.2862
```

```
var(wblake$Length[wblake$Age==3])
```

```
## [1] 411.6679
```

As the variance is constant then it produces a null plot for the standard deviation compared to ages.

```
#####THANKYOU#####
```